ANKIT YADAV

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Course	College/University	Year	CGPA/%
M.Tech. Computer Science	Indian Statistical Institute Kolkata	2025-27	61
M.Sc. Mathematics	Indian Institute of Technology (IIT), Bombay	2022-24	6.07
B.Sc. Mathematics	Indira Gandhi University, Rewari	2018-21	69.84

ACHIEVEMENTS/FELLOWSHIPS

- Secured All India Rank 88 in IIT JAM Mathematics.
- Achieved 90.35 percentile in IIT JAM Statistics.
- Merit-cum-Means Scholarship during M.Sc. at IIT Bombay.

RELEVANT COURSES

M.Tech. Discrete Mathematics; Automata Theory; Operating System; Computer Organization; Randomized & Approximation Algorithms; Quantum Computing; Statistical Methods;

M.Sc. Linear Algebra; Basic Algebra; Algebra-I; Optimization; Introduction to Fourier Analysis

RESEARCH EXPERIENCE

Analysis of CRYSTALS-Dilithium, qTESLA and FALCON
Guides: Dr. Sartaj Ul Hasan & Dr. Sumit Kumar Pandey, IIT Jammu |DRDO Funded

- Reproduced published solutions for specific instances of the LWE challenge and evaluated their solvability under relaxed boundary conditions.
- Experimentally validated key-recovery attack on reduced-parameter variants of FALCON and Module-FALCON.

PROJECTS

AI-assisted Quantum Error Correction Simulation

- Simulated a distance-3 rotated surface code using Stim to model Pauli noise across 3 measurement rounds, simulating a 26-qubit system to generate fault datasets.
- Benchmarked neural decoding against Minimum-Weight Perfect Matching (MWPM) via PyMatching, evaluating logical-error suppression across physical error rates ($p \in [0.005, 0.020]$) and code distances ($d = 3, 5, 7$).
- Trained a PyTorch MLP decoder on 135,000 fault samples to map raw syndrome vectors to logical observables, achieving 89.82% accuracy and a 76.68% F1-score by optimizing with weighted cross-entropy to mitigate class imbalance.

Quantum Data Management

- Studied why standard Database management systems cannot perfectly be used for Quantum Data.
- Used SQLite to execute quantum gates, bypassing the 2^n dense vector memory wall for sparse algorithms.
- Benchmarked performance of the RDBMS-based sparse approach against traditional dense simulator approach by traditional quantum simulators (e.g., Qiskit).

RELEVANT SKILLS

- **Programming Languages**: Python, C, SQL
- **Python Libraries**: SciPy, PyTorch, Qiskit, Stim, Numpy, Matplotlib, Pandas

SEMINARS AND WORKSHOPS

- Attended ISEA webinar held at ISI under the aegis of Ministry of Electronics and Information Technology.
- Attended Topics in Advanced Areas of Cryptology(TAAC) workshop held at ISI under the aegis of DRDO.
- Attended Theory day at Arvind Raghunathan Center for Theoretical Computer Science at IIT Madras.
- Attended National Symposium on Quantum Information and Foundations organized by PAMU, ISI.